Field	Details
Document Title	API Test Case Execution Document
Project Title	API Testing using Postman
Assignment	Assignment 3 – API Testing
Submitted By	Kaleemuddin Mohammed
Application Under Test	ReqRes Public API (https://reqres.in)
Testing Tool	Postman
Objective	To perform API testing covering GET, POST, PUT, and DELETE methods, including validation of status codes, response body, response time, and handling of positive, negative, and invalid scenarios.
Submission Date	01-Apr-2026

Test Cases Execution

Test Case ID: TC_001
Module: GET Users
Test Scenario: Retrieve list of users with valid request
Test Environment:

- Tool: Postman
- API: https://reqres.in
- Method: GET

Execution Steps

1. Open Postman application.
2. Select GET request method.
3. Enter URL: https://reqres.in/api/users?page=2
4. Go to Headers tab.
5. Add header:
    - Key: x-api-key
    - Value: (Actual API key)
6. Click on Send button.
7. Observe response status code, response body, and response time.

Expected Result

- Status code should be **200 OK**
- Response body should contain list of users
- Response should include fields like page, total, and data
- Response time should be within acceptable limit

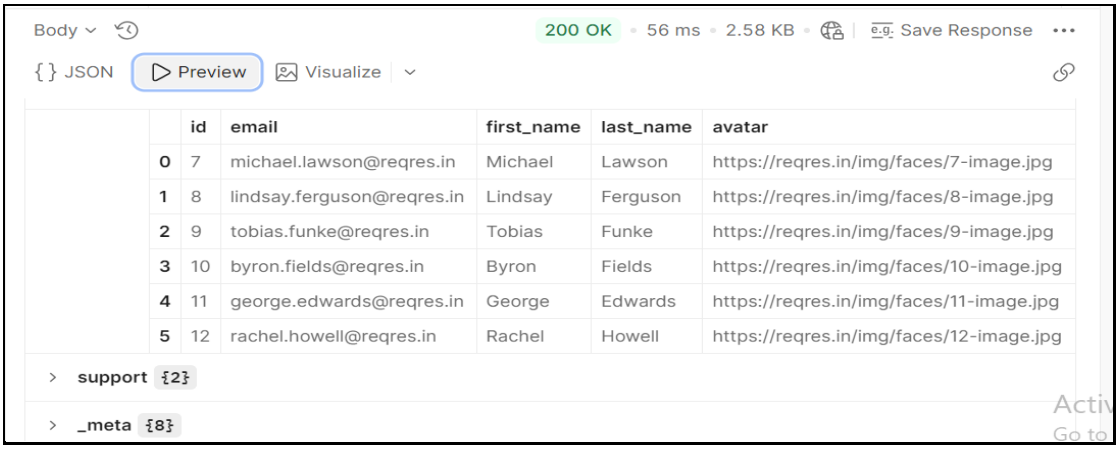
Actual Result

- Status code: 200 OK
- User list displayed successfully
- Response structure is valid

Status

PASS

Execution Evidence



	id	email	first_name	last_name	avatar
0	7	michael.lawson@reqres.in	Michael	Lawson	https://reqres.in/img/faces/7-image.jpg
1	8	lindsay.ferguson@reqres.in	Lindsay	Ferguson	https://reqres.in/img/faces/8-image.jpg
2	9	tobias.funke@reqres.in	Tobias	Funke	https://reqres.in/img/faces/9-image.jpg
3	10	byron.fields@reqres.in	Byron	Fields	https://reqres.in/img/faces/10-image.jpg
4	11	george.edwards@reqres.in	George	Edwards	https://reqres.in/img/faces/11-image.jpg
5	12	rachel.howell@reqres.in	Rachel	Howell	https://reqres.in/img/faces/12-image.jpg

Test Case ID: TC_002
Module: GET Users
Test Scenario: Validate response structure for user list

Test Environment

- Tool: Postman
- API: https://reqres.in
- Method: GET

Execution Steps

1. Open Postman application.
2. Select GET request method.
3. Enter URL: https://reqres.in/api/users?page=2
4. Go to Headers tab.
5. Add header:
    - Key: x-api-key
    - Value: (Actual API key)
6. Click on Send button.
7. Observe the response body.
8. Verify the presence of required fields in the response.

Expected Result

- Status code should be **200 OK**
- Response body should contain:
    - page
    - per_page
    - total
    - total_pages
    - data (array of users)

Actual Result

- Status code: 200 OK
- Response contains all required fields
- Data array with user details is present

Status

PASS

Execution Evidence

My Collection > **ReqRes API Testing**

Save | Share

GET https://reqres.in/api/users?page=2 **Send**

Docs | Params | Authorization | Headers 7 | Body | Scripts | Settings | Cookies

Body | 200 OK • 56 ms • 2.58 KB | Save Response

{ } JSON | Preview | Visualize


```
{
    "page": 2,
    "per_page": 6,
    "total": 12,
    "total_pages": 2,
    "data": [
        {
            "id": 7,
            "email": "michael.lawson@reqres.in",
            "first_name": "Michael",
            "last_name": "Lawson",
            "avatar": "https://reqres.in/img/faces/7-image.jpg"
        },
        {
            "id": 8,
            "email": "lindsay.ferguson@reqres.in",
            "first_name": "Lindsay",
            "last_name": "Ferguson",
            "avatar": "https://reqres.in/img/faces/8-image.jpg"
        },
        {
            "id": 9,
            "email": "tobias.funke@reqres.in"
```

Test Case ID: TC_003
Module: GET Users
Test Scenario: Retrieve users without API key

Test Environment

- Tool: Postman
- API: https://reqres.in
- Method: GET

Execution Steps

1. Open Postman application.
2. Select GET request method.
3. Enter URL: https://reqres.in/api/users?page=2
4. Go to Headers tab.
5. **Remove the x-api-key header** (or do not add it).
6. Click on Send button.
7. Observe the response status and message.

Expected Result

- Status code should be **401 Unauthorized**
- Response should show error message indicating missing API key

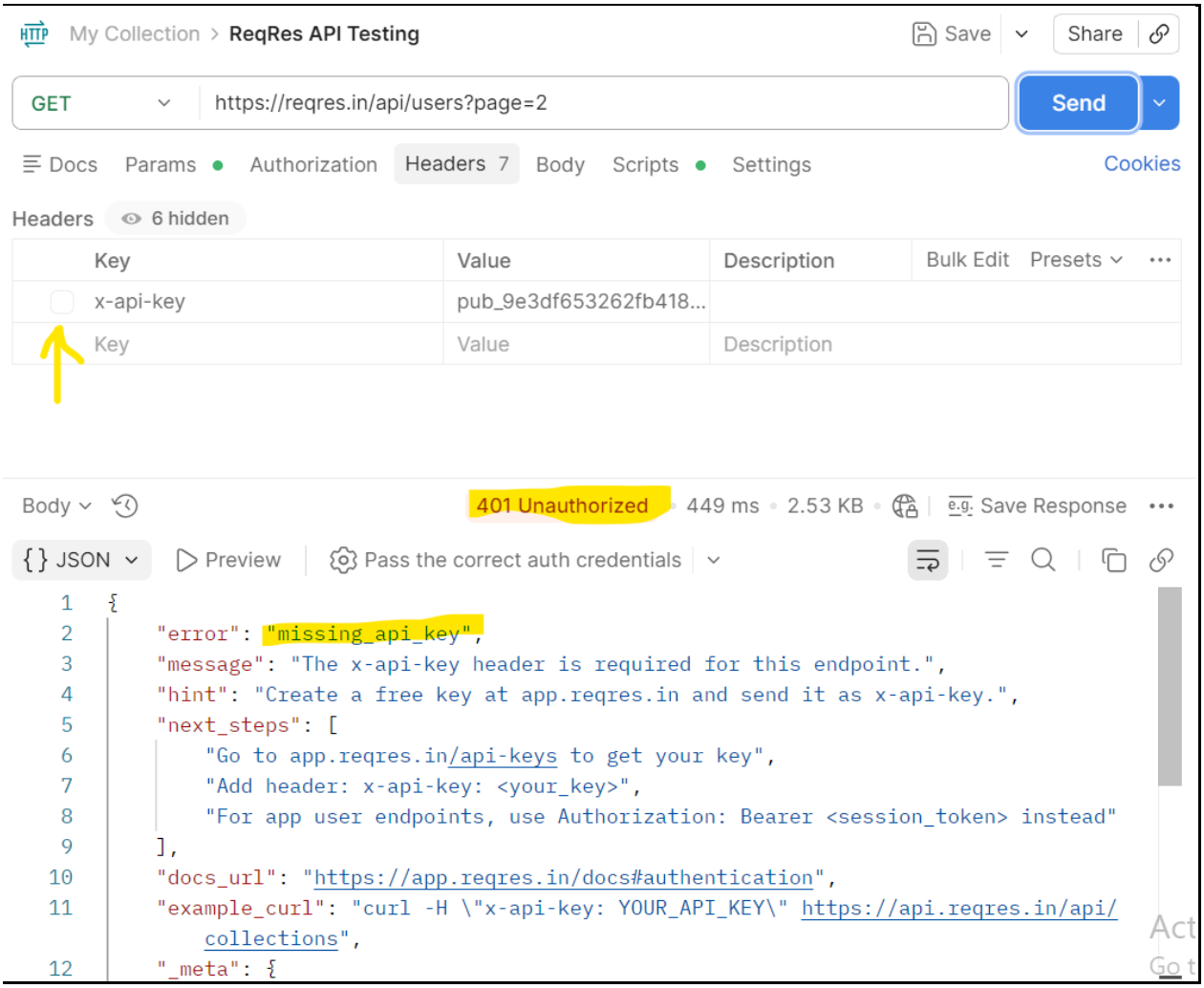
---

Actual Result

- Status code: 401 Unauthorized
- Error message: missing_api_key

Status
PASS

Execution Evidence



Test Case ID: TC_004
Module: GET Users
Test Scenario: Retrieve single user with valid ID

Test Environment

- Tool: Postman
- API: https://reqres.in
- Method: GET

Execution Steps

1. Open Postman application.
2. Select GET request method.
3. Enter URL: https://reqres.in/api/users/2
4. Go to Headers tab.
5. Add header:
    - Key: x-api-key
    - Value: (Actual API key)
6. Click on Send button.
7. Observe the response body and status code.

Expected Result

- Status code should be **200 OK**
- Response should contain details of the user with ID = 2
- Fields like id, email, first_name, last_name should be present

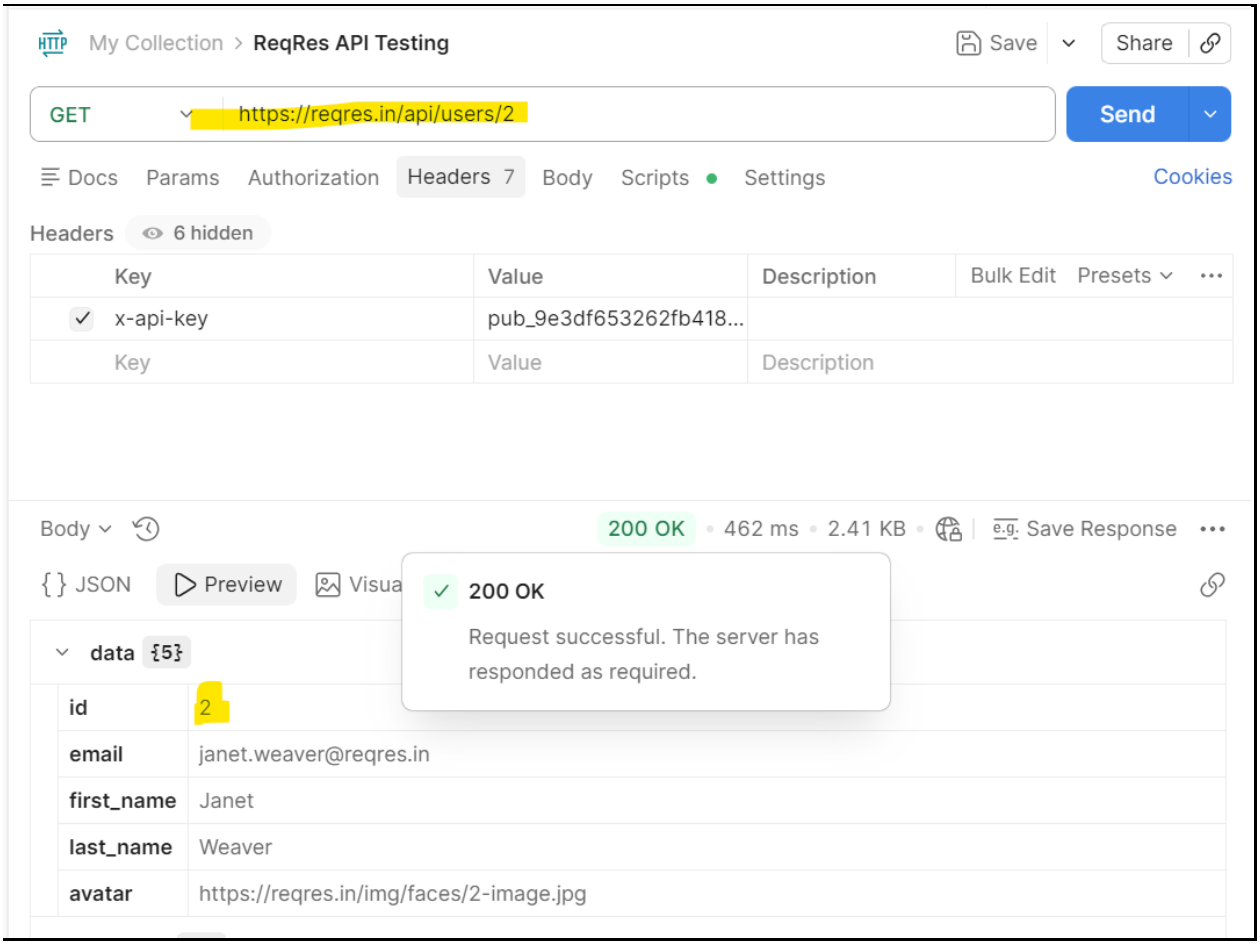
---

Actual Result

- Status code: 200 OK
- User details are returned correctly
- Required fields are present

Status
PASS

Execution Evidence



Test Case ID: TC_005
Module: GET Users
Test Scenario: Retrieve single user with invalid ID

Test Environment

- Tool: Postman
- API: https://reqres.in
- Method: GET

Execution Steps

1. Open Postman application.
2. Select GET request method.
3. Enter URL: https://reqres.in/api/users/23
4. Go to Headers tab.
5. Add header:
    - Key: x-api-key
    - Value: (Actual API key)
6. Click on Send button.
7. Observe the response status and body.

Expected Result

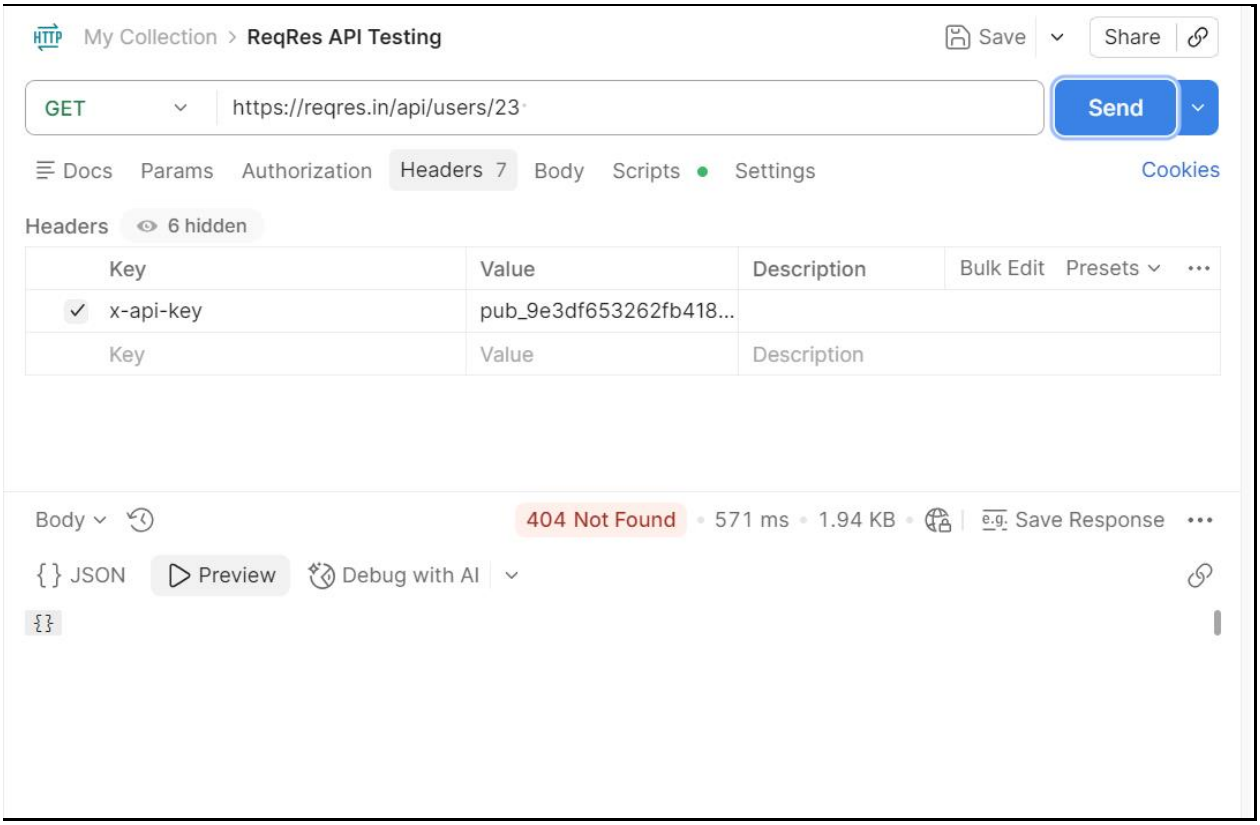
- Status code should be **404 Not Found**
- Response body should be empty or indicate user not found

Actual Result

- Status code: 404 Not Found
- No user data returned

Status
PASS

Execution Evidence



Test Case ID: TC_006
Module: POST User
Test Scenario: Create user with valid data

Test Environment

- Tool: Postman
- API: https://reqres.in
- Method: POST

Execution Steps

1. Open Postman application.
2. Select POST request method.
3. Enter URL: https://reqres.in/api/users
4. Go to Headers tab.
5. Add headers:
    - Key: x-api-key
    - Value: (your API key)
    - Key: Content-Type
    - Value: application/json
6. Go to **Body → raw → JSON**
7. Enter request body:

```
JSON

{
  "name": "Kaleemuddin",
  "job": "QA Engineer"
}
```

8. Click on Send button.
9. Observe the response status and response body.

Expected Result

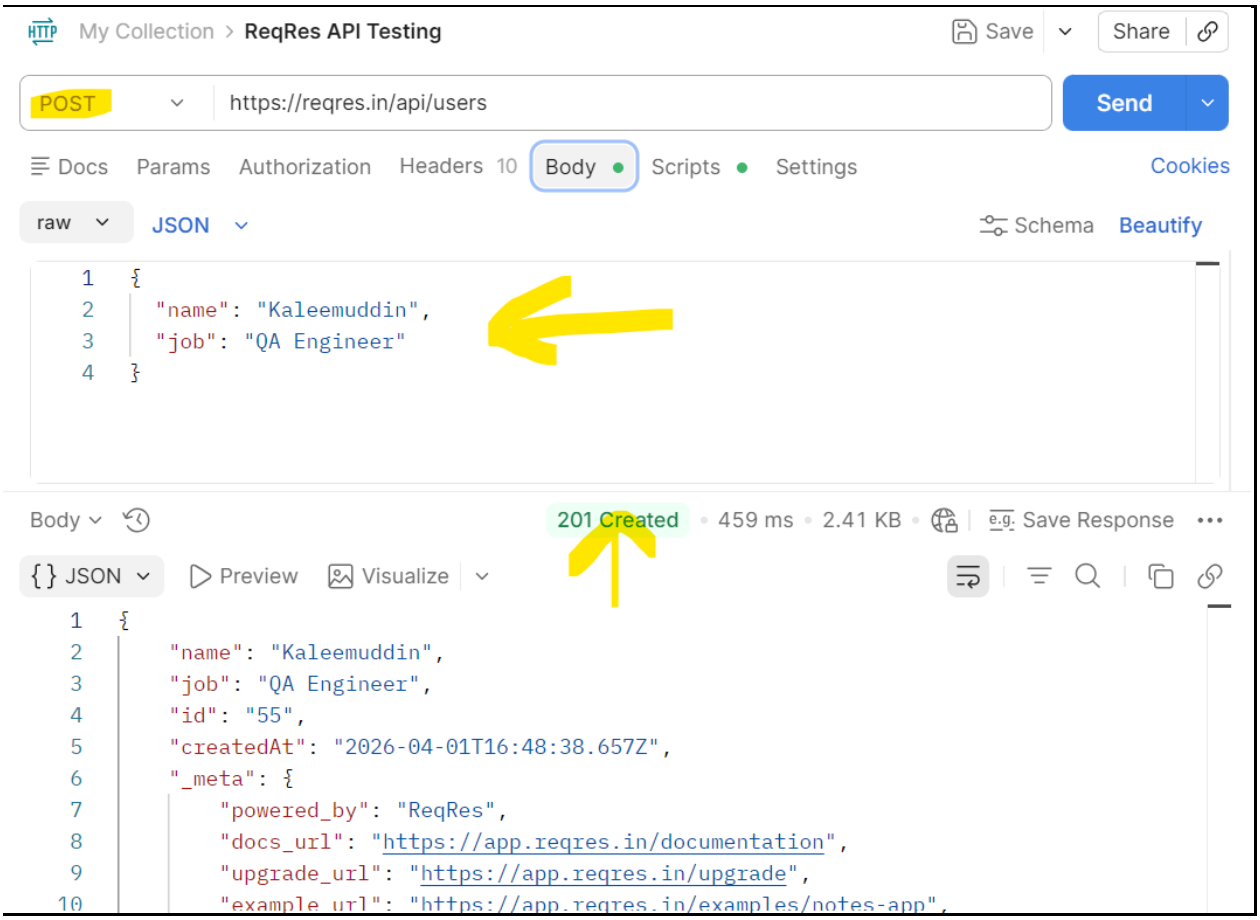
- Status code should be **201 Created**
- Response should contain:
    - name
    - job
    - id
    - createdAt

Actual Result

- Status code: 201 Created
- User created successfully
- id and createdAt fields are returned

Status PASS

Execution Evidence



Test Case ID: TC_007
Module: POST User
Test Scenario: Validate POST response body

Test Environment

- Tool: Postman
- API: https://reqres.in
- Method: POST

Execution Steps

1. Open Postman application.
2. Select POST request method.
3. Enter URL: https://reqres.in/api/users
4. Go to Headers tab.
5. Add headers:
    - x-api-key → (Actual API key)
    - Content-Type → application/json
6. Go to Body → raw → JSON
7. Enter:

```
{
  "name": "John",
  "job": "Tester"
}
```

8. Click Send
9. Observe response body carefully

Expected Result

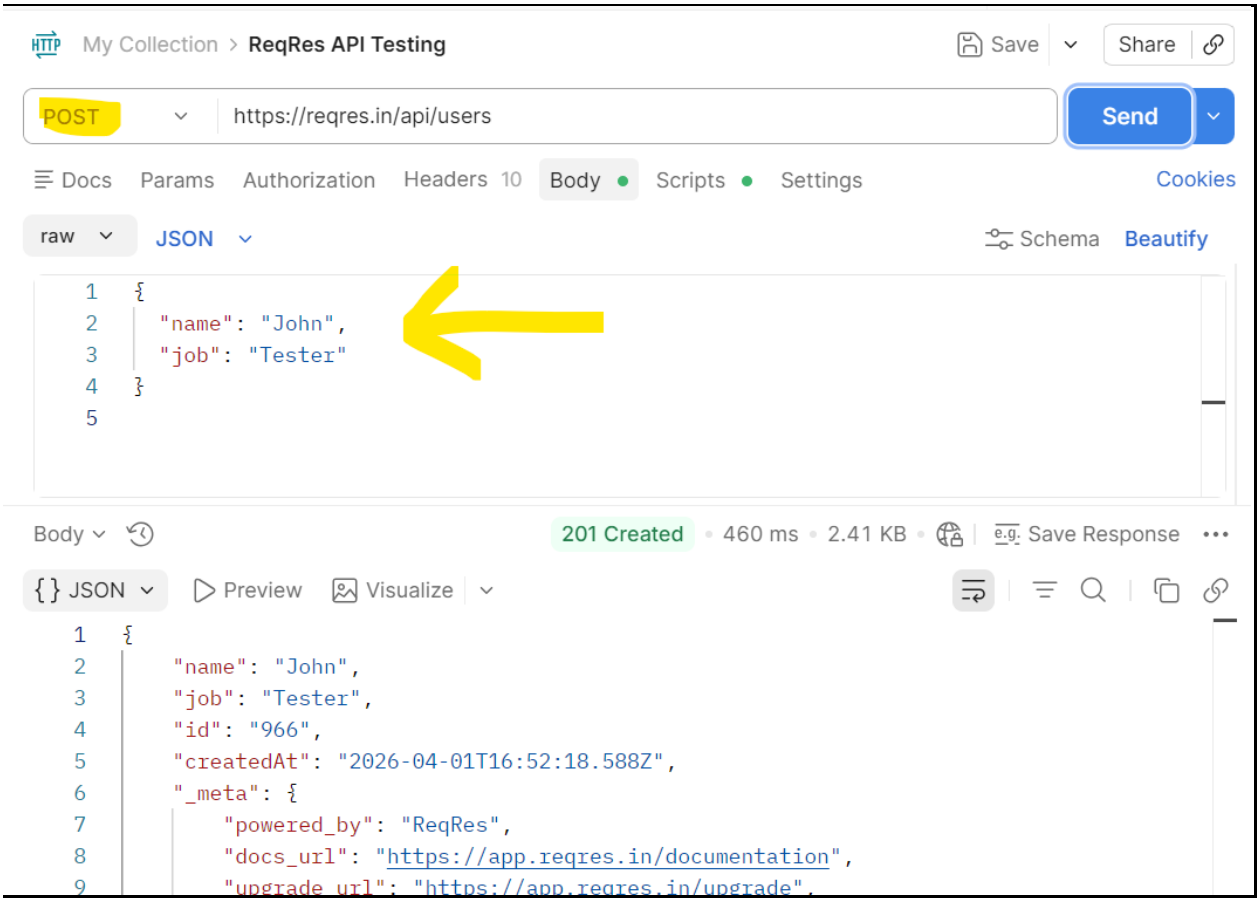
- Status code should be **201 Created**
- Response should contain:
    - name = "John"
    - job = "Tester"
    - id (generated)
    - createdAt (timestamp)

Actual Result

- Status code: 201 Created
- name and job match request
- id and createdAt fields are present

Status: PASS

Execution Evidence



Test Case ID: TC_008
Module: POST User
Test Scenario: Create user with empty request body

Test Environment

- Tool: Postman
- API: https://reqres.in
- Method: POST

Execution Steps

1. Open Postman application.
2. Select POST request method.
3. Enter URL: https://reqres.in/api/users
4. Go to Headers tab.
5. Add headers:
    - x-api-key → (Actual API key)
    - Content-Type → application/json
6. Go to Body → raw → JSON
7. Enter empty body:

   { }

8. Click Send
9. Observe response status and body

Expected Result

- Ideally, API should return **400 Bad Request** for empty input
- OR handle the request gracefully (since this is a mock API)

Actual Result

- Status code: **201 Created** (ReqRes behavior)
- API still returns id and createdAt

Status: PASS

POST https://reqres.in/api/users Send

Docs Params Authorization Headers 10 Body Scripts Settings Cookies

raw JSON Schema Beautify

```
1 {}
```

Body 201 Created 491 ms 2.38 KB Save Response

{} JSON Preview Visualize


```
1 {
2     "id": "190",
3     "createdAt": "2026-04-01T16:54:56.521Z",
4     "_meta": {
5         "powered_by": "ReqRes",
```

Test Case ID: TC_009
Module: POST User
Test Scenario: Create user with missing field

Test Environment

- Tool: Postman
- API: https://reqres.in
- Method: POST

Execution Steps

1. Open Postman application.
2. Select POST request method.
3. Enter URL: https://reqres.in/api/users
4. Go to Headers tab.
5. Add headers:
    - x-api-key → (Actual API key)
    - Content-Type → application/json
6. Go to Body → raw → JSON
7. Enter request body:

```
{
  "name": "OnlyName"
}
```

8. Click Send
9. Observe response status and body

Expected Result

- Ideally, API should return **400 Bad Request** for missing required field
- OR handle request with partial data

Actual Result

- Status code: **201 Created**
- API accepts request and returns id and createdAt

Status: FAIL

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POST https://reqres.in/api/users **Send**

Docs | Params | Authorization | Headers 10 | Body | Scripts | Settings | Cookie

raw | JSON | Schema | Beautify

```
1 {
2   "name": "OnlyName"
3 }
4
```

Body | 201 Created • 459 ms • 2.39 KB | Save Response

{} JSON | Preview | Visualize

```
1 {
2   "name": "OnlyName",
3   "id": "971",
4   "createdAt": "2026-04-01T17:00:04.328Z",
5   "_meta": {
```

Test Case ID: TC_010
Module: POST User
Test Scenario: Validate POST response time

Test Environment

- Tool: Postman
- API: https://reqres.in
- Method: POST

Execution Steps

1. Open Postman application.
2. Select POST request method.
3. Enter URL: https://reqres.in/api/users
4. Go to Headers tab.
5. Add headers:
    - x-api-key → (Actual API key)
    - Content-Type → application/json
6. Go to Body → raw → JSON
7. Enter request body:

```
{
  "name": "API",
  "job": "Automation"
}
```

8. Click Send
9. Observe **response time** (top right in Postman)

Expected Result

- Status code should be **201 Created**
- Response time should be **less than 2000 ms**

Actual Result

- Status code: 201 Created
- Response time: (e.g., 300–600 ms observed)

Status: PASS

My Collection > **ReqRes API Testing**

Save | Share

POST https://reqres.in/api/users Send

Docs | Params | Authorization | Headers 10 | Body | Scripts | Settings | Cookies

raw | JSON | Schema | Beautify

```
1 {
2   "name": "API",
3   "job": "Automation"
4 }
5
```

Body | 201 Created | 441 ms | 2.4 KB | Save Response

{} JSON | Preview | Visualize

```
1 {
2   "name": "API",
3   "job": "Automation",
4   "id": "304",
```

Test Case ID: TC_011
Module: PUT User
Test Scenario: Update user with valid data

Test Environment

- Tool: Postman
- API: https://reqres.in
- Method: PUT

Execution Steps

1. Open Postman application.
2. Select **PUT** request method.
3. Enter URL:

https://reqres.in/api/users/2

4. Go to Headers tab.
5. Add headers:
    - x-api-key → (Actual API key)
    - Content-Type → application/json
6. Go to **Body → raw → JSON**
7. Enter request body:

```
{
  "name": "Updated",
  "job": "Senior QA"
}
```

8. Click Send
9. Observe response body and status

Expected Result

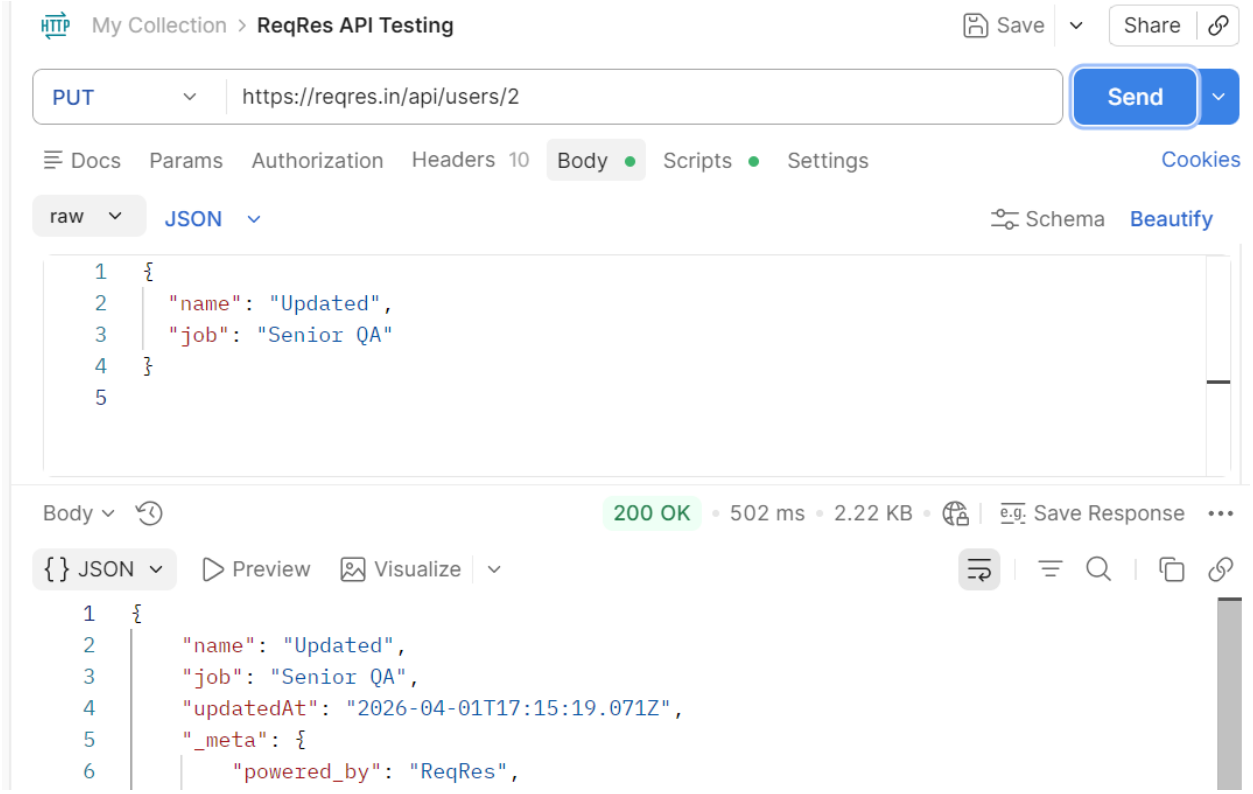
- Status code should be **200 OK**
- Response should contain:
    - name = "Updated"
    - job = "Senior QA"
    - updatedAt (timestamp)

Actual Result

- Status code: 200 OK
- User details updated successfully
- updatedAt field is present

Status : PASS

Execution Evidence



Test Case ID: TC_012
Module: PUT User
Test Scenario: Validate updated response fields

Test Environment

- Tool: Postman
- API: https://reqres.in
- Method: PUT

Execution Steps

1. Open Postman application.
2. Select **PUT** request method.
3. Enter URL:

https://reqres.in/api/users/2

4. Go to Headers tab.
5. Add headers:
    - x-api-key → (Actual API key)
    - Content-Type → application/json
6. Go to **Body → raw → JSON**
7. Enter request body:

```
{
  "name": "Maria",
  "job": "Lead"
}
```

8. Click Send
9. Observe response body carefully

Expected Result

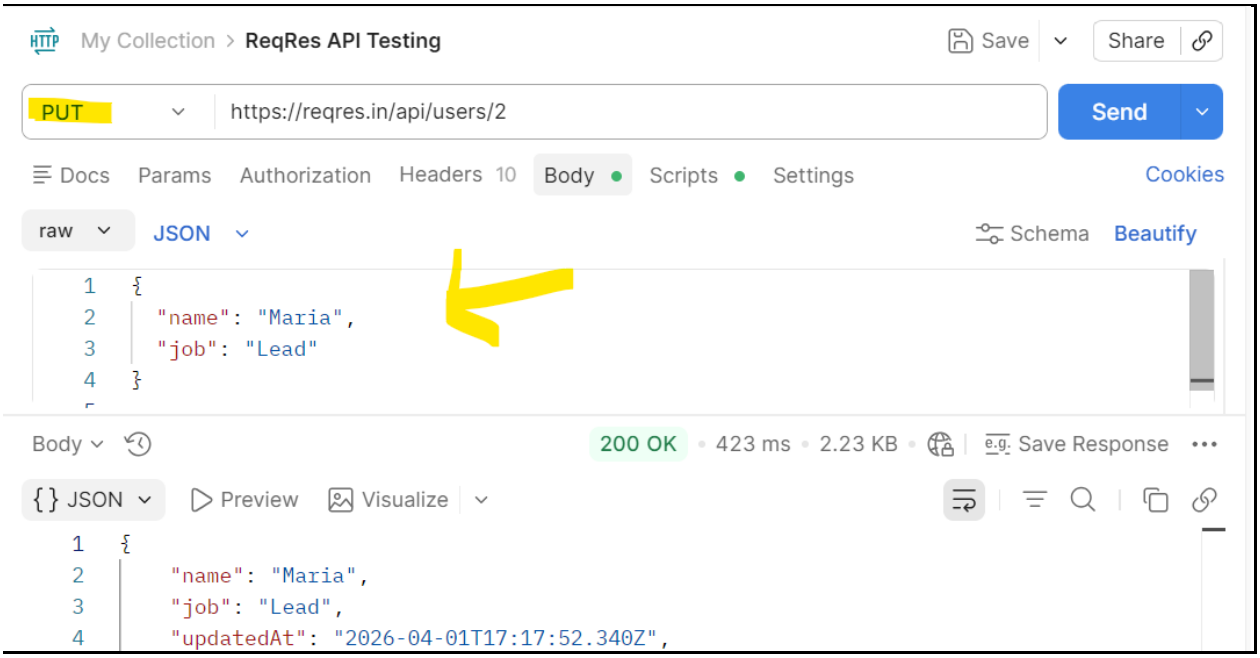
- Status code should be **200 OK**
- Response should contain:
    - name = "Maria"
    - job = "Lead"
    - updatedAt field present
- Returned values should match the request

Actual Result

- Status code: 200 OK
- name and job match the request
- updatedAt field is present

Status: PASS

Execution Evidence



Test Case ID: TC_013
Module: PUT User
Test Scenario: Update user with invalid data types

Test Environment

- Tool: Postman
- API: https://reqres.in
- Method: PUT

Execution Steps

1. Open Postman application.
2. Select **PUT** request method.
3. Enter URL: https://reqres.in/api/users/2
4. Go to Headers tab.
5. Add headers:
    - x-api-key → (Actual API key)
    - Content-Type → application/json
6. Go to **Body → raw → JSON**
7. Enter request body:

```
{
  "name": 123,
  "job": true
}
```

8. Click Send
9. Observe response status and body

Expected Result

- Ideally, API should reject invalid data types and return **400 Bad Request**
- Or handle the request according to mock API behavior

Actual Result

- Status code: **200 OK**
- API accepts the payload and returns updatedAt field

Status : FAIL

My Collection > **ReqRes API Testing**

Save | Share

PUT https://reqres.in/api/users/2 **Send**

Docs | Params | Authorization | Headers 10 | Body | Scripts | Settings | Cookies

raw | JSON | Schema | Beautify

```
1 {
2   "name": 123,
3   "job": true
4 }
```

Body | 200 OK • 522 ms • 2.22 KB | Save Response

{ } JSON | Preview | Visualize

```
1 {
2     "name": 123,
3     "job": true,
4     "updatedAt": "2026-04-01T17:22:16.194Z",
5     "_meta": {
6         "powered_by": "ReqRes",
```

Test Case ID: TC_014
Module: DELETE User
Test Scenario: Delete user with valid ID

Test Environment

- Tool: Postman
- API: https://reqres.in
- Method: DELETE

Execution Steps

1. Open Postman application.
2. Select **DELETE** request method.
3. Enter URL:

`https://reqres.in/api/users/2`

4. Go to Headers tab.
5. Add header:
   - x-api-key → (Actual API key)
6. Click Send
7. Observe response status and body

Expected Result

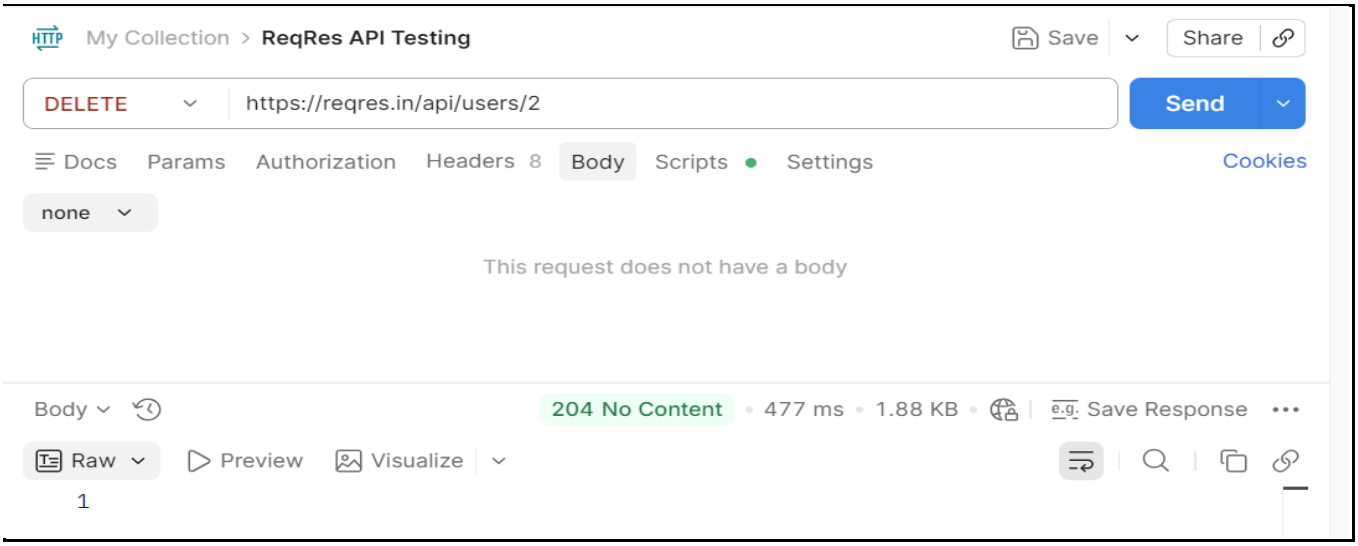
- Status code should be **204 No Content**
- Response body should be empty

Actual Result

- Status code: 204 No Content
- Response body is empty

Status: PASS

Execution Evidence



Test Case ID: TC_015
Module: DELETE User
Test Scenario: Delete user without API key (Unauthorized access)

Test Environment

- Tool: Postman
- API: https://reqres.in
- Method: DELETE

Execution Steps

1. Open Postman application.
2. Select **DELETE** request method.
3. Enter URL:

https://reqres.in/api/users/2

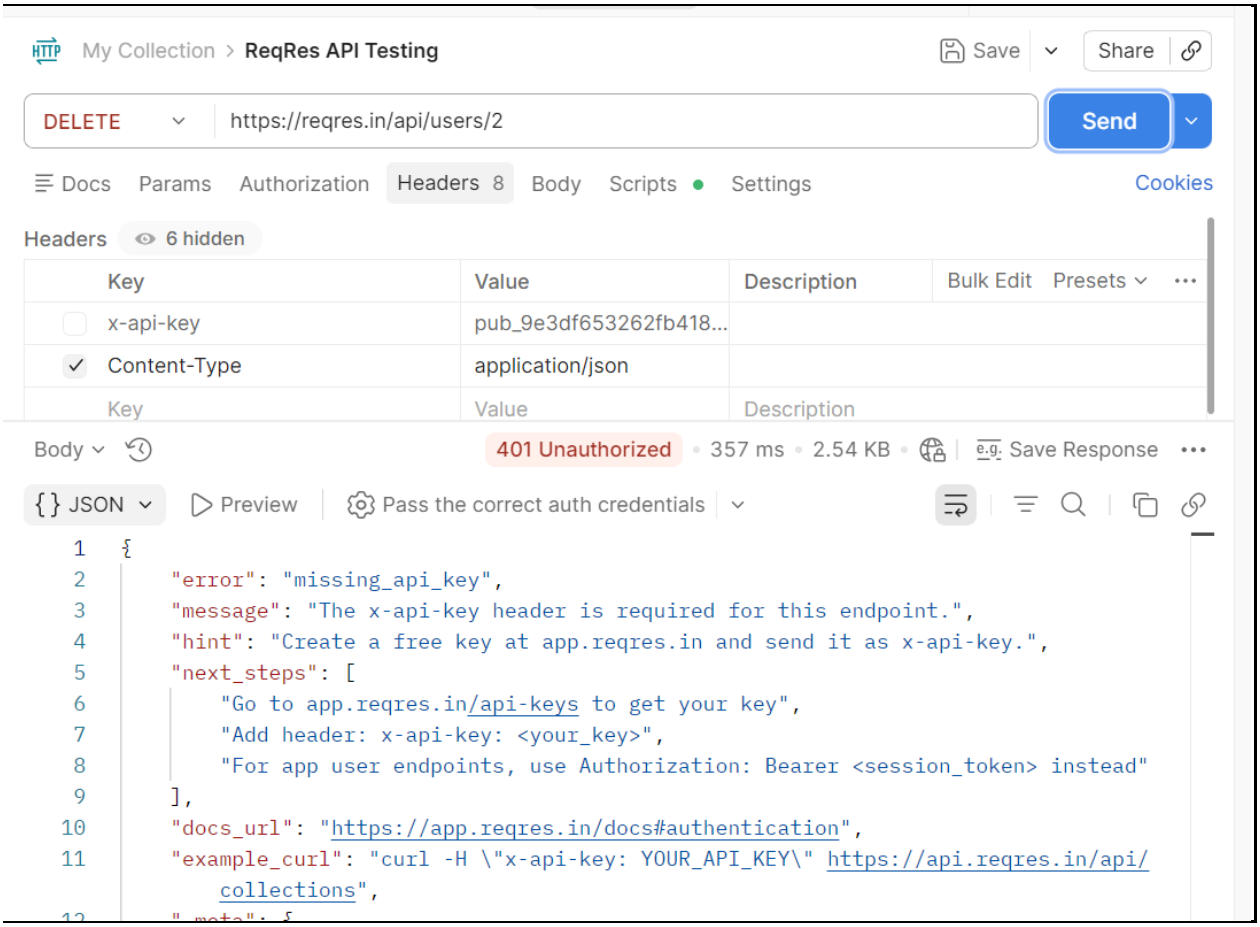
4. Go to Headers tab.
5. **Remove the x-api-key header** (or do not add it).
6. Ensure Body is set to NONE
7. Click Send
8. Observe response status and message

Expected Result

- Status code should be **401 Unauthorized**
- Response should indicate missing API key

Actual Result

- Status code: 401 Unauthorized
- Error message: missing_api_key

Status PASS

Execution Evidence